



RAJAMAHENDRI INSTITUTE OF ENGINEERING AND TECHNOLOGY

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

Gesture-Controlled wheel Chair And Home Automation System By using MEMS sensor (MPU60501)

PRESENTED BY

**Under the guidance of,
N.CHANDRA SEKHAR M.tech
Assistant Professor**

B.HEMANTH KUMAR	(21MD1A0406)
K.SHYAM	(21MD1A0433)
D.T.LAXMIKNATH	(21MD1A0416)

PROJECT TITLE

**Gesture-Controlled wheel Chair And Home
Automation System**

By using MEMS sensor (MPU60501)

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Abstract

With the rise of smart technologies gesture-based systems are growing in use. This project uses a MEMS sensor to capture hand gestures and convert them into commands to control both a wheelchair and home appliances. The goal is to create a simple, low-cost solution that is easy to use and helps improve user accuracy.

Introduction

With the rapid evolution of smart technologies, gesture-controlled systems are becoming increasingly popular in robotics and home automation this project aims to design and implement a system that uses an MEMS SENSOR to give hand gestures and convert these signals into control commands for robot movement and appliance automation. The primary goal is to offer a cost-effective, user-friendly solution for enhanced control and efficiency

Literature Survey

In the existing systems most traditional wheelchairs rely on joystick based control, which requires manual operation and dexterity. This limits accessibility for users with severe physical disabilities.

Although some modern systems have introduced voice commands and limited automation features they still pose challenges such as high cost complex operation and lack of integration with home automation.

Existing Features

- Sensor-Based Gesture Recognition
- Binary Device Control
- Limited Motion Control
- Appliance Automation
- Minimal Integration

Proposed System

- Gesture-Based Control with high efficiency
- Seamless Integration
- Multi-Directional Movement.
- Appliance Automation
- Real-Time Processing

Block Diagram

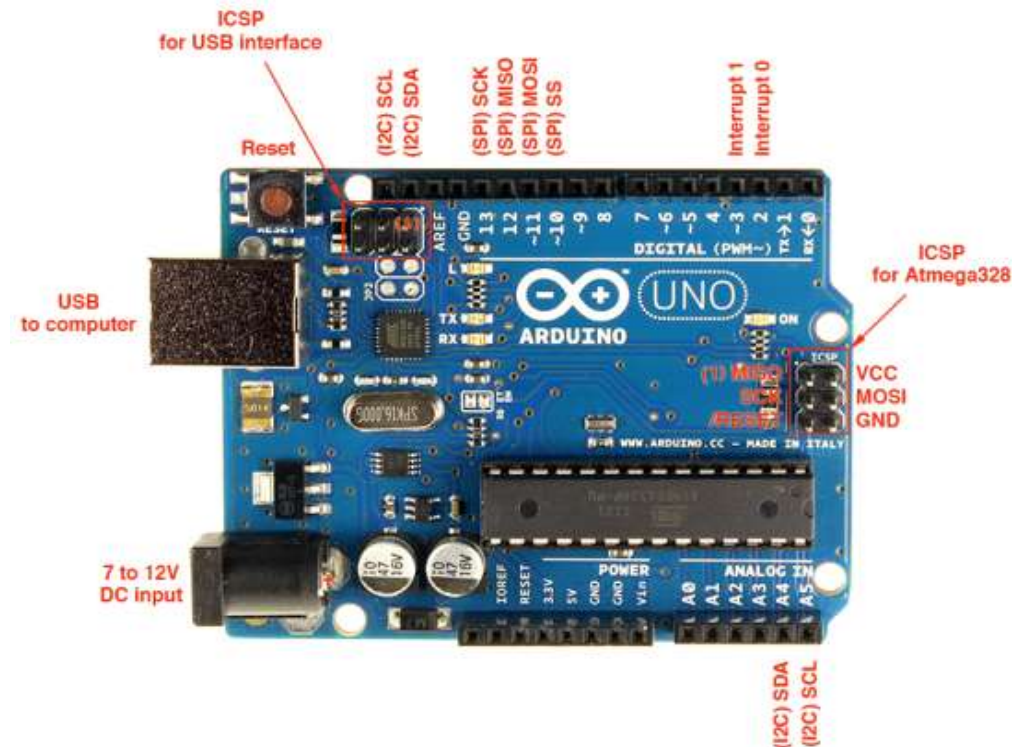


COMPONENTS

- Arduino Uno
- MEMS sensor
- IR Sensor
- L298N Motor Driver
- DC Motors

ARDUINO UNO BOARD

The Arduino Uno is a microcontroller board based on the ATmega328P. It serves as the brain of the system, processing inputs from the IR sensor and generating outputs to control motors and appliances. It supports both analog and digital inputs, making it suitable for controlling various components in real time



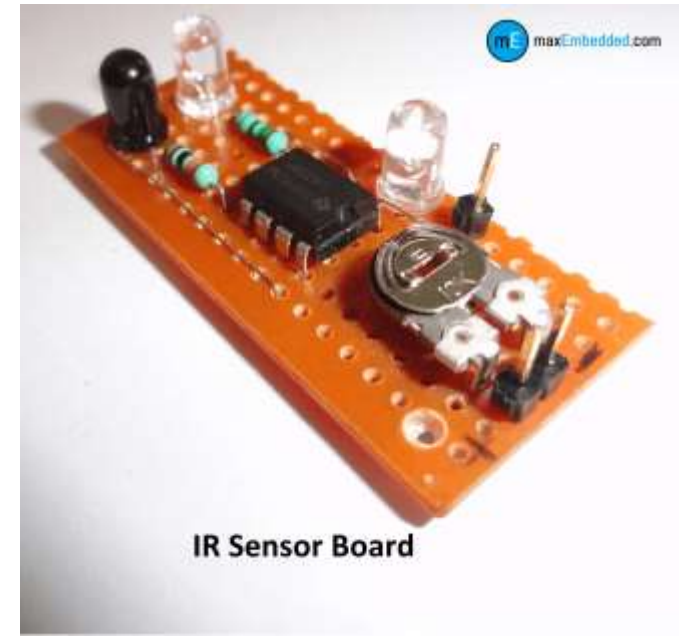
MEMS SENSOR(MPU60501)

The **MPU6051** is a **MEMS (Micro-Electro-Mechanical Systems)** sensor that combines a **3-axis gyroscope** and a **3-axis accelerometer** on a single chip. It is widely used in robotics, gesture-controlled devices, and motion tracking applications



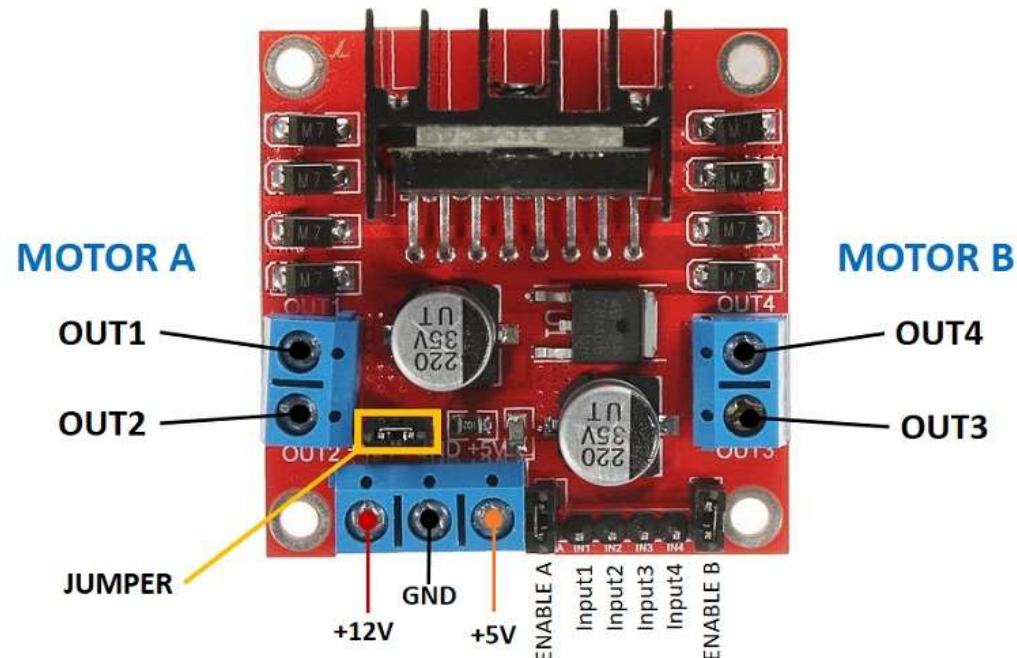
IR SENSOR

This project offers an easy-to-use and cost-effective solution for people with mobility challenges. It allows users to control their wheelchair and home devices using simple gestures. With future upgrades, the system can become even more helpful and widely usable.



L293D Motor Driver

This motor driver is used to control the direction and speed of DC motors. It acts as an interface between the Arduino and the motors, ensuring smooth bidirectional movement based on the gesture inputs. It supports multiple motors and provides overcurrent protection

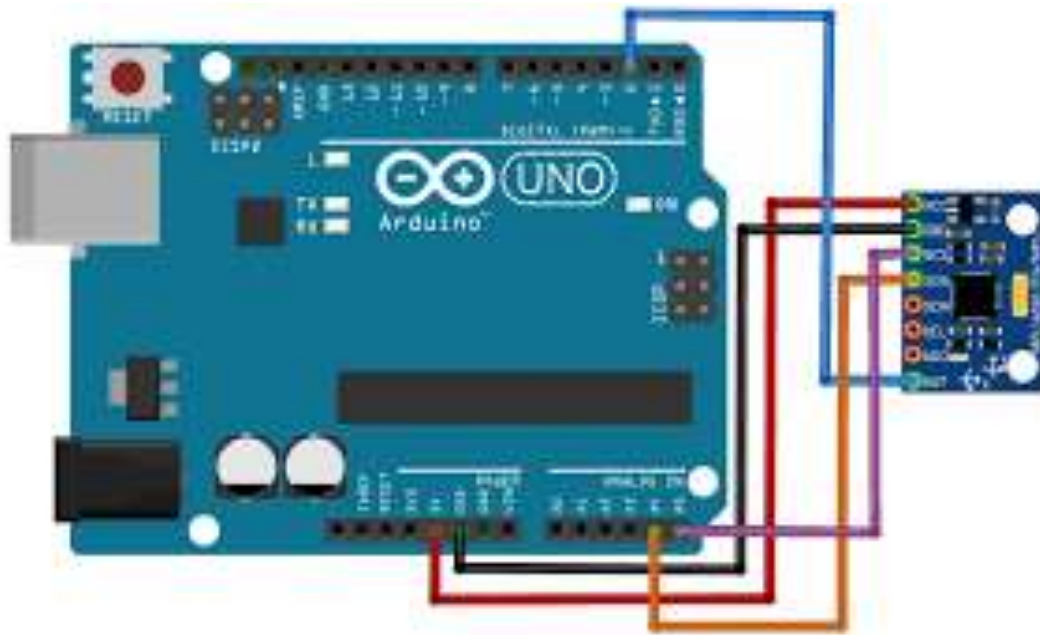


DC Motors

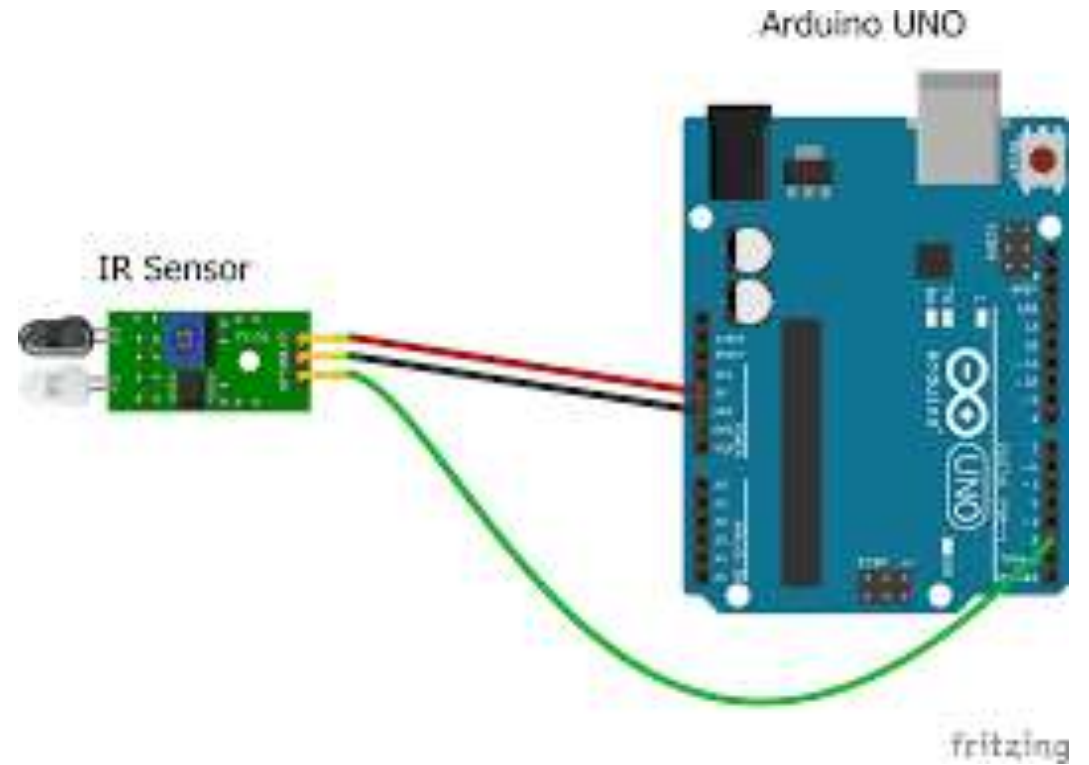
The DC motors drive the robot's movement in different directions. The system uses two motors for left and right wheels, enabling control for forward, backward, left, and right motion. The speed and direction of these motors are controlled by the motor driver



Sensor interfacing with Arduino UNO



MEMS SENSOR WITH ARDUINO UNO



IR SENSOR WITH ARDUINO UNO

Working

- The MEMS sensor detects hand or head gestures made by the user.
- These gestures are sent as signals to the Arduino Uno microcontroller.
- Arduino processes the signals and sends commands to the L293D motor driver.
- The motor driver controls the movement of the wheelchair in different directions.
- An IR sensor is used to control home appliances like fans or lights remotely.
- A buzzer provides safety alerts when obstacles are detected or errors occur.

SOFTWARE IMPLIMENTATION

Arduino is a prototype platform (open-source) based on an easy-to-use hardware and software. It consists of a circuit board, which can be programmed (referred to as a microcontroller) and a ready-made software called Arduino IDE (Integrated Development Environment), which is used to write and upload the computer code to the physical board

ADVANTAGES

- Hands-Free Control
- Enhanced Mobility for Disabled Users
- Real-Time Response
- Seamless Integration
- Low Cost and Easy Maintenance
- User-Friendly Design
- Energy Efficiency
- Scalability

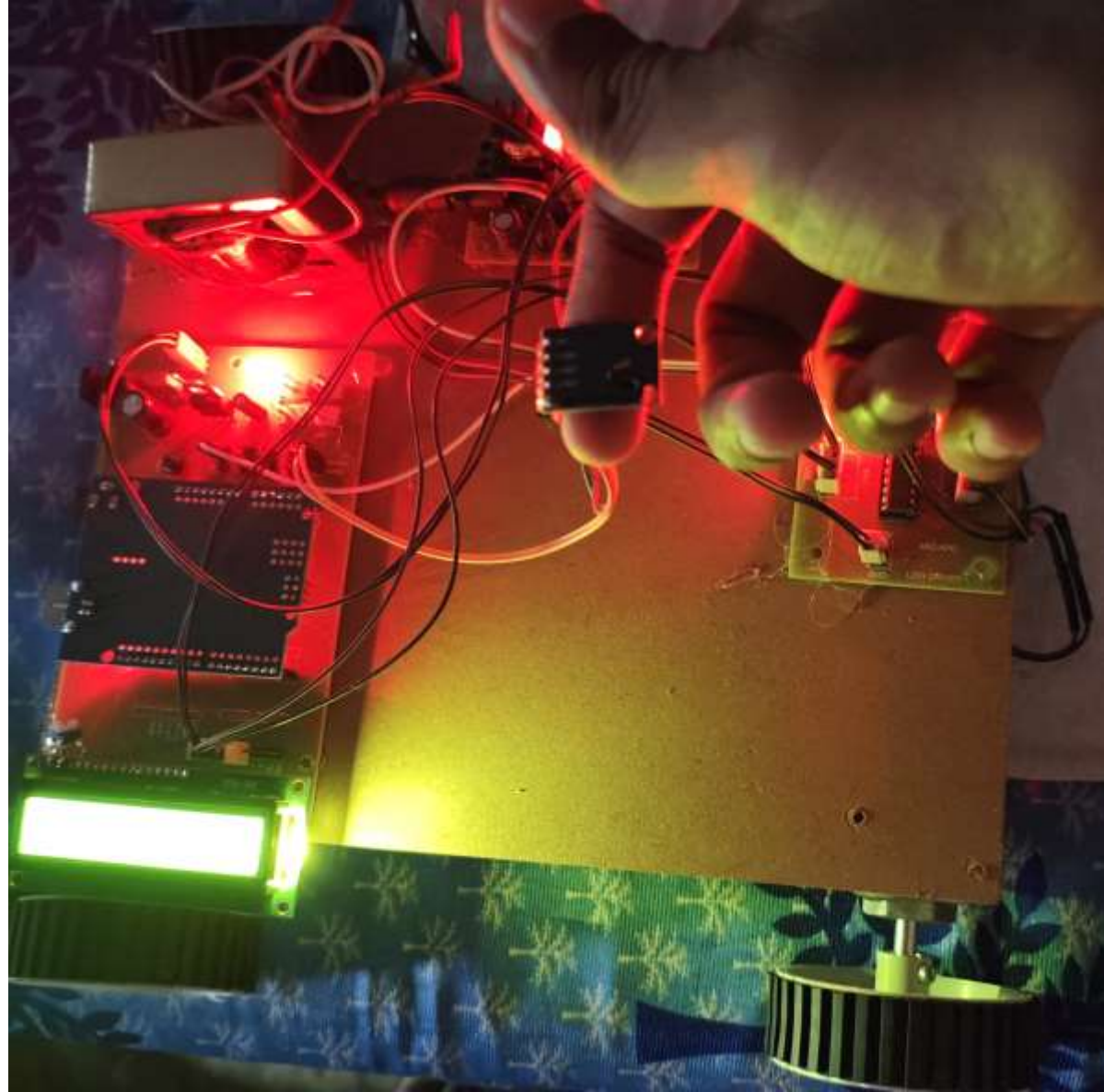
FUTURE SCOPE

- Can be improved with voice control for easier use.
- Wireless control using Bluetooth or Wi-Fi can be added.
- Obstacle detection can be included for safety.
- Gestures can be customized for each user.
- Health monitoring sensors can be added.
- Solar power can be used to save battery

Real World Applications

- Assisting individuals with spinal cord injuries or paralysis
- Used in hospitals and rehabilitation centers for patient mobility
- Elderly care for improved independence
- Smart homes for integrated control of devices
- Used in disaster response where remote control is essential
- Enhances accessibility in public spaces

RESULT



CONCLUSION

This project helps people control a wheelchair using hand gestures. It is easy to use low-cost and helpful for disabled users. With some improvements it can become even more useful in the future

THANK YOU